SESSION 1.1 – INTRODUCTION TO AI

ACTIVITY WITH ANSWERS

Activity: Students will act as a machine and discuss the four goals of AI, namely thinking/acting humanly and thinking/acting rationally, to understand their differences.

Answer:

Artificial Intelligence can be understood through four major approaches. Each approach describes a different goal for an intelligent system: to think like a human, act like a human, think logically, or act in the best possible way. The four goals are explained below.

AI Goal	Meaning	Example
1. Thinking Humanly	The machine tries to imitate the way humans actually think. It focuses on human thought processes such as learning, remembering, reasoning and solving problems.	Example: An AI tutor that studies how a student understands a problem and adapts its explanation like a human teacher.
2. Acting Humanly	The machine tries to behave like a human. The emphasis is on producing human-like responses and actions, even if the internal thinking process is different.	Example: A chatbot that can communicate naturally with a person. The Turing Test is closely related to this approach.
3. Thinking Rationally	The machine tries to reason correctly by using logic, rules and valid conclusions. The goal is to think according to the principles of rational or logical reasoning.	Example: An expert system that applies logical rules to facts in order to reach a conclusion.
4. Acting Rationally	The machine chooses the action that is most likely to achieve the best result according to its goal and the information available. This is the rational-agent approach used widely in modern AI.	Example: A self-driving car selecting the safest route and reacting to traffic conditions to reach its destination safely.

Main Difference: “Humanly” approaches focus on copying human thinking or behavior, whereas “rationally” approaches focus on correct reasoning or choosing the best action. “Thinking” refers to the internal reasoning process, while “acting” refers to the external behavior of the system.

Conclusion: The four goals show that AI can be designed either to imitate humans or to behave rationally. Modern AI systems often follow the acting rationally approach because it helps an intelligent agent select actions that best achieve its objectives.

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